Large time behavior of reaction-diffusion equations with Bessel generators

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Abstract

We investigate explosion in finite time of one-dimensional semilinear equations of the form

$$\frac{\partial u_t}{\partial t}(x) = \frac{1}{2}\frac{\partial^2 u_t}{\partial x^2}(x) + \frac{\varphi'(x)}{\varphi(x)}\frac{\partial u_t}{\partial x}(x) - \frac{a}{x^2}u_t(x) + u_t^{1+\beta}(x)$$

with initial value $\phi \geq 0$, where $\varphi \in \mathcal{C}^2(\mathbb{R})$ is positive and $a \geq 0$, $\beta > 0$ are constants. In the free case $a = 0$ we provide conditions on φ under which any positive nontrivial solution is non-global. In the case $a > 0$ and $\varphi(x) = x^{\mu+1/2}$, $\mu \in \mathbb{R}$, which includes in the special case $\mu = -1/2$ the equation

$$\frac{\partial u_t}{\partial t}(x) = \frac{1}{2}\frac{\partial^2 u_t}{\partial x^2}(x) - \frac{a}{x^2}u_t(x) + u_t^{1+\beta}(x),$$

we use the Feynman-Kac formula for Bessel processes to give conditions on the equation parameters ensuring finite-time blowup and existence of nontrivial positive global solutions.

Key words: Semilinear PDEs, Bessel processes, Feynman-Kac representation, critical exponent, finite time blow-up, global solution.

Mathematics Subject Classification: 60H30, 35K55, 35K57, 35B35.

1 Introduction

Consider a semilinear PDE of the form

$$\begin{cases} \dfrac{\partial u_t}{\partial t}(x) = Au_t(x) - V(x)u_t(x) + G(u_t(x)), \quad t > 0, \\ \\ u_0(x) = \phi(x), \quad x \in \mathbb{R}^N, \end{cases}$$

where A is the generator of a nice Markov process in $\mathbb{R}^N$ and $V(x)$, $x \in \mathbb{R}^N$, is a nonnegative potential, $G(z) \geq 0$ is a nonlinear term which is locally Lipschitz, say of the form $G(z) = z^{1+\beta}$ for some $\beta > 0$, and the initial value $u_0(x) = \phi(x)$ is bounded and nonnegative. In this setting, a positive solution $u_t(x)$ will either exist up to a

finite life span τ_f, and in this case $\|u_t\|_\infty$ explodes as t approaches τ_f from below, or exists globally in the sense that $\|u_t\|_\infty < \infty$ for all $t > 0$. In the former case it is said that $u_t(x)$ blows up in finite time, and in the latter that $u_t(x)$ is a global solution.

It is well-known that, when $V = 0$ and $A = \Delta$ is the Laplacian in the N-dimensional Euclidean space, the ratio $2/N$ rules out the asymptotic growth of $u_t(x)$. More precisely, if $\beta \leq 2/N$ then, apart from 0, there are no nonnegative global solutions, whereas if $\beta > 2/N$, the equation admits nontrivial positive global solutions.

When $V(x) > 0$ is constant one can prove under mild conditions on A that $u_t(x)$ must be global if we choose ϕ appropriately small. This follows from the fact that, since $V > 0$ and e^{tA} is a contraction for every $t > 0$, $\|e^{-tV}e^{tA}\varphi\|_\infty$ decays exponentially fast as $t \to \infty$, and this suffices to conclude that $u_t(x)$ is global if $\phi(x)$ decays quickly enough to 0 as $\|x\|$ goes to infinity.

The case when $V(x) \geq 0$ is non-constant is less clear and has been studied both in the analytic and probabilistic literature. For example, critical exponents for the finite time blow-up of the semilinear equation

$$\begin{cases} \dfrac{\partial u_t}{\partial t}(x) = \dfrac{1}{2}\Delta u_t(x) - V(x)u_t(x) + u_t^{1+\beta}(x), \quad t > 0, \\ \\ u_0(x) = \phi(x), \quad x \in \mathbb{R}^N, \end{cases}$$

in dimensions $N \geq 3$, where $\phi(x) \geq 0$ and $V(x)$ is bounded above (or below) by $a/(1 + |x|^b)$, $x \in \mathbb{R}^N$, with $a > 0$ and $b \in [2, \infty)$, have been studied in [11], [14], [15]. In [7] we treated this type of problem using heat kernel estimates and the Feynman-Kac representation, again for $N \geq 3$, but including in the critical value $b = 2$. It turns out that, if $N \geq 3$ and

$$0 \leq V(x) \leq \frac{a}{1 + \|x\|^b}, \qquad x \in \mathbb{R}^N,$$

where $a > 0$ and $b > 2$, then $u_t(x)$ blows up in finite time for all $0 < \beta < 2/N$. On the contrary, if

$$V(x) \geq \frac{a}{1 + \|x\|^b}, \qquad x \in \mathbb{R}^N,$$

with $a > 0$ and $0 \leq b < 2$, then $u_t(x)$ can be global for any $\beta > 0$ provided $\phi(x)$ decays sufficiently fast to 0 as x goes to infinity. For the "critical" value $b = 2$, blowup occurs when $0 < \beta < \beta_*(a)$, where the upper bound $\beta_*(a)$ depends rather sensitively on a; see [6], [7], [11]. Thus, substracting a nontrivial potential to the diffusion term

in our equation may be a delicate question.

The unbounded potential $V(x) = a/|x|^2$ has been considered when $N \geq 3$ in [5], [1], and [2], where it is shown that (1.6) admits a unique critical exponent $\beta(a) < 2/N$, given by

$$\beta(a) = \frac{4}{2 + N + \sqrt{8a + (N-2)^2}}. \tag{1.1}$$

Namely, if $V(x) = a|x|^{-2}$, then no global nontrivial solution of (1.6) exists if $\beta < \beta(a)$, whereas global solutions exist if $\beta > \beta(a)$.

As noted above, most of the existing literature on the blowup of semilinear PDEs with potential deals with the case $N \geq 3$. In this paper we investigate the one-dimensional case of semilinear equations of the form

$$\begin{cases} \dfrac{\partial u_t}{\partial t}(x) = \dfrac{1}{2}\dfrac{\partial^2 u_t}{\partial x^2}(x) + \dfrac{\varphi'(x)}{\varphi(x)}\dfrac{\partial u_t}{\partial x}(x) - \dfrac{a}{x^2}u_t(x) + u_t^{1+\beta}(x), \quad t > 0, \\ \\ u_0(x) = \phi(x) \geq 0, \quad x > 0, \end{cases} \tag{1.2}$$

where $\beta > 0$ and $a \geq 0$ are constants, and ϕ is a non identically vanishing measurable function on $\mathbb{R}$. The case $N = 1$ has several features that make it different from that of $N \geq 3$, in particular the potential $V(x) = a/x^2$, $a > 0$, is not integrable around 0 and the underlying Brownian motion returns to 0 with probability one.

We proceed as follows. In Section 2 we first consider the "free case" $a = 0$ and $\mathcal{C}^2$ functions $\varphi \in L^2(\mathbb{R})$ such that φ'/φ is bounded. Using Jensen's inequality, together with the fact that $\mu(dx) = \varphi^2(x)\,dx$ is an invariant measure of the semigroup $(T_t)_{t\in\mathbb{R}_+}$ with generator

$$L^\varphi f(x) := \frac{1}{2}\frac{\partial^2 f}{\partial x^2}(x) + \frac{\varphi'(x)}{\varphi(x)}\frac{\partial f}{\partial x}(x),$$

we arrive at the conclusion that $\|u_t\|_\infty$ explodes in finite time.

In the remaining sections 3, 4, 5 and 6 we consider the case $a \geq 0$ with

$$\varphi(x) = x^{\mu+1/2}, \quad x \in \mathbb{R},$$

where $\mu \in \mathbb{R}$, which renders the equation

$$\begin{cases} \dfrac{\partial u_t}{\partial t}(x) = L_\mu u_t(x) - \dfrac{a}{x^2}u_t(x) + u_t^{1+\beta}(x), \quad t > 0, \\ \\ u_0(x) = \phi(x), \quad x > 0, \end{cases} \tag{1.3}$$

where

$$L_\mu = \frac{1}{2}\frac{\partial^2}{\partial x^2} + \frac{2\mu+1}{2x}\frac{\partial}{\partial x}$$

is the generator of the Bessel process of index $\mu \in \mathbb{R}$, and $\beta > 0$ is a constant. This case falls apart from the setting outlined above, including in the case of $a = 0$, therefore we need to resort to other techniques. Our main tools here are the Feynman-Kac representation of (1.3), as well as certain bounds established in Sections 3 and 4 for the heat kernel and for the conditional moments of the Bessel generator perturbed by $V(x)$. This approach also allows us to deal with a class of convex increasing nonlinearities $G(z)$, and with certain time-dependent nonlinear terms of the form $t^\zeta G(u_t(x))$.

Next in Section 5 we prove that for $\mu \in \mathbb{R}$, if $a > 0$ and

$$\beta < \frac{2+\mu-\sqrt{\mu^2+2a}}{2+\mu+\sqrt{\mu^2+2a}}, \tag{1.4}$$

then (1.3) possesses no nontrivial positive global solutions, cf. Corollary 5.3. We also deal with a semilinear equation whose nonlinear term is of the form $t^\zeta G(u_t(x))$, where $\zeta \geq 0$ is a constant and $G(z)$ is a positive increasing convex function satisfying certain growth conditions, see (5.3) below.

In Section 6 we show that for $\mu \in \mathbb{R}$ and $a > 0$, (1.3) admits a global positive nontrivial solution when

$$\beta > \frac{2}{2+\mu+\sqrt{\mu^2+2a}}, \tag{1.5}$$

cf. Theorem 6.2. When $\mu > -1$ and $a = 0$ we find $2/d := (1+\mu)^{-1}$ as the critical exponent for explosion of (1.3), where $d = 2 + 2\mu$ denotes the dimension of the underlying Bessel process with parameter $\mu \in \mathbb{R}$. Notice that when $\mu \geq 0$, (1.4) and (1.5) recover the critical exponent $(1+\mu)^{-1}$ as a tends to 0, while in case $\mu < 0$ the equation exhibits a discontinuous behavior and the bounds (1.4) and (1.5) do not apply when $a = 0$. This is due to the fact that the Bessel process almost surely does not return to 0 when $\mu \geq 0$, and returns to 0 in finite time with strictly positive probability when $\mu < 0$. In particular when $\mu = -1/2$ and $a > 0$, i.e. $d = 1$, this shows that the equation

$$\frac{\partial u_t}{\partial t}(x) = \frac{1}{2}\frac{\partial^2 u_t}{\partial x^2}(x) - \frac{a}{x^2}u_t(x) + u_t^{1+\beta}(x), \tag{1.6}$$

(with a nontrivial positive u_0) blows up in finite time for

$$\beta < \frac{3-\sqrt{1+8a}}{3+\sqrt{1+8a}} < \beta(a),$$

and admits a global solution when

$$\beta > \frac{4}{3+\sqrt{1+8a}} = \beta(a),$$

which partly extends (1.1) to the case $N = 1$. As noted above, when a tends to 0 the above thresholds do not recover the critical exponent 2, which is obtained separately in Corollary 5.3 and Theorem 6.2 below.

2 The free case

In this section we consider the case $a = 0$ and prove that (1.2) blows-up in finite time when $\varphi : \mathbb{R} \to (0, \infty)$ is in $L^2(\mathbb{R})$, of class $\mathcal{C}^2$, and such that the function

$$x \longmapsto \frac{\varphi'(x)}{\varphi(x)}, \quad x \in \mathbb{R},$$

is bounded. First, let us explain our method in the particular case

$$\varphi(x) = e^{-x^2/2}, \qquad x \in \mathbb{R},$$

in which (1.2) becomes

$$\frac{\partial u_t}{\partial t}(x) = \frac{1}{2}\frac{\partial^2 u_t}{\partial x^2}(x) - x\frac{\partial u_t}{\partial x}(x) + u_t^{1+\beta}(x), \qquad t > 0,$$

where

$$L^\varphi = \frac{1}{2}\frac{\partial^2}{\partial x^2} - x\frac{\partial}{\partial x}$$

is the generator of the Ornstein-Uhlenbeck process with semigroup $(T_t)_{t\in\mathbb{R}_+}$. Then the semigroup $(Q_t)_{t\in\mathbb{R}_+}$ defined by

$$Q_t f(x) := e^{-x^2/2} T_t\left(e^{x^2/2} f(x)\right), \qquad t \in \mathbb{R}_+,$$

is the Markov semigroup corresponding to the harmonic oscillator, a real-valued Gauss-Markov process having generator

$$H = \frac{1}{2}\frac{\partial^2}{\partial x^2} - \frac{x^2}{2} + \frac{1}{2}. \tag{2.1}$$

Since Q_t can be written as

$$Q_t f(x) = \int_{-\infty}^{\infty} K_t(x,y) f(y)\, dy,$$

where the kernel $K_t(x,y)$ is given by Mehler's formula, it is easy to verify that for any positive $\phi \in L^2(\mu)$ and all sufficiently large t we have

$$Q_t \phi(x) \geq c\kappa(\beta t)^{-1/\beta}, \qquad x \in \mathbb{R}, \tag{2.2}$$

for some constant $c > 0$, and

$$\kappa = \inf_{t \geq 1} \inf_{|x|,|y|<1} K_t(x,y) > 0,$$

provided ϕ does not identically vanish. The inequality (2.2) above ensures finite-time blowup of the solution of

$$\begin{cases} \dfrac{\partial u_t}{\partial t}(x) = H u_t(x) + u_t^{1+\beta}(x), \\ \\ u_0(x) = \phi(x), \qquad x \in \mathbb{R}, \quad t > 0, \end{cases}$$

see [8] and [9]. From here we can infer that the norm $\|u_t\|_\infty$ of the solution $u_t(x)$ to (1.2) with $a = 0$ explodes in finite time.

In the general case, when the function $\varphi(x)$ is not specified, one cannot expect to know explicitly the transition densities of $(Q_t)_{t \in \mathbb{R}_+}$, however we obtain the following result.

Theorem 2.1 *Let $\varphi \in L^2(\mathbb{R}) \cap \mathcal{C}^2(\mathbb{R})$ be a positive function such that*

$$x \mapsto \frac{\varphi'(x)}{\varphi(x)}, \quad x \in \mathbb{R},$$

is bounded. Let $\mu(dx) := \varphi^2(x)\, dx$, and let G be a positive convex function such that

$$G(z) \geq c z^{1+\beta} \text{ for all } z \geq 0,$$

where $\beta > 0$ and $c > 0$ is constant. Then the norm $\|u_t\|_{L^1(\mu)}$ of any positive solution of

$$\begin{cases} \dfrac{\partial u_t}{\partial t}(x) = L^\varphi u_t(x) + G(u_t(x)), \\ \\ u_0(x) = \phi(x) \geq 0, \qquad x \in \mathbb{R}, \quad t > 0, \end{cases} \tag{2.3}$$

blows up in finite time, provided its initial value $\phi \geq 0$ has the form

$$\phi(x) = \frac{h(x)}{\varphi(x)} \geq 0, \quad x \in \mathbb{R}, \tag{2.4}$$

for some positive nontrivial $h \in \mathcal{C}^2(\mathbb{R}) \bigcap L^2(\mu)$.

Proof. By a classical comparison argument, see e.g. Lemma 3.1 in [6], it suffices to consider the case $G(z) = z^{1+\beta}$, $z \geq 0$. Writing again $(T_t)_{t\in\mathbb{R}_+}$ for the semigroup with generator L^φ, which now is given by

$$L^\varphi := \frac{1}{2}\frac{\partial^2}{\partial x^2} + \frac{\varphi'(x)}{\varphi(x)}\frac{\partial}{\partial x},$$

we get

$$u_t(x) = T_t\phi(x) + \int_0^t T_{t-s}u_s^{1+\beta}(x)\, ds, \qquad x \in \mathbb{R}, \quad t \geq 0. \tag{2.5}$$

By multiplying the above equation by $\varphi(x)$, letting $w_t(x) := \varphi(x)u_t(x)$ and using (2.4), we obtain

$$w_t(x) = \varphi(x)T_t\left(\frac{h}{\varphi}\right)(x) + \int_0^t \varphi(x)T_{t-s}\left(w_s^{1+\beta}\varphi^{-\beta-1}\right)(x)\, ds.$$

Notice that

$$f \mapsto Q_t^\varphi f := \varphi T_t\left(\frac{f}{\varphi}\right), \quad t \geq 0, \tag{2.6}$$

defines a semigroup $(Q_t^\varphi)_{t\in\mathbb{R}_+}$ of bounded linear operators on $L^2(\mu)$ with infinitesimal generator given by

$$H^\varphi f = \frac{1}{2}\frac{\partial^2}{\partial x}f - \frac{1}{2}\frac{\varphi''}{\varphi}f, \qquad f \in \mathcal{C}^2(\mathbb{R}), \tag{2.7}$$

and that

$$w_t(x) = (Q_t^\varphi h)(x) + \int_0^t Q_{t-s}^\varphi\left(w_s^{1+\beta}(\cdot)\varphi^{-\beta}\right)(x)\, ds. \tag{2.8}$$

Let now

$$\mathcal{E}(f) = \langle f, \varphi\rangle_{L^2(\mathbb{R})}, \quad f \in L^2(\mu).$$

Since $\tilde{h} := \varphi^2$ satisfies

$$\frac{1}{2}\frac{\partial^2\tilde{h}}{\partial x^2}(x) - \frac{\partial}{\partial x}\left(\tilde{h}(x)\frac{\varphi'(x)}{\varphi(x)}\right) = 0,$$

it turns out that $\mu(dx) = \varphi^2(x)\, dx$ is an invariant probability measure of $(T_t)_{t\in\mathbb{R}_+}$ up to normalization, and it follows from (2.8) that

$$\mathcal{E}(w_t) = \mathcal{E}(h) + \int_0^t \mathcal{E}\left(w_s^{1+\beta}\varphi^{-\beta}\right)\, ds. \tag{2.9}$$

Now,

$$\begin{aligned}\mathcal{E}\left(w_s^{1+\beta}\varphi^{-\beta}\right) &= \|\varphi\|_{L^2}^2 \int_{-\infty}^{\infty} \left(\frac{w_s^{1+\beta}}{\varphi^{1+\beta}}\right)(x) \frac{\varphi^2(x)}{\|\varphi\|_{L^2}}\, dx \\ &\geq \frac{\|\varphi\|_{L^2}^2}{\|\varphi\|_{L^2}^{2+2\beta}} \left(\int_{\mathbb{R}} \frac{w_s(x)}{\varphi(x)} \varphi^2(x)\, dx\right)^{1+\beta} \\ &= \|\varphi\|_{L^2}^{-2\beta} (\mathcal{E}(w_s))^{1+\beta},\end{aligned}$$

where we used Jensen's inequality. Hence from (2.9) we have

$$\mathcal{E}(w_t) \geq \mathcal{E}(w_0) + \|\varphi\|_{L^2}^{-2\beta} \int_0^t (\mathcal{E}(w_s))^{1+\beta} ds.$$

Let now $y(t)$ be the solution to the ordinary differential equation

$$y(t) = \mathcal{E}(w_0) + \|\varphi\|_{L^2}^{-2\beta} \int_0^t y^{1+\beta}(s) ds. \tag{2.10}$$

Since $\mathcal{E}(w_t)$ is a supersolution of (2.10) and $y(t)$ explodes at time

$$t_{h,\varphi} = \frac{\|\varphi\|_{L^2}^{2\beta}}{\beta \left(\int_{-\infty}^{\infty} h(x)\varphi(x)\, dx\right)^{\beta}},$$

it follows that

$$\mathcal{E}(w_t) = \|u_t\|_{L^1(\mu)} = +\infty, \qquad t \geq t_{h,\varphi}.$$

□

From the inequality

$$\int \varphi^2(x) u_t(x)\, dx \;\leq\; \|u_t\|_{\infty} \int_{\mathbb{R}} \varphi^2(x)\, dx, \qquad t > 0,$$

we get the following corollary of Theorem 2.1.

Corollary 2.2 *Let $\beta > 0$, and let $\phi \geq 0$ be of the form (2.4). Then under the assumptions of Theorem 2.1, the solution of* (2.3) *blows up in finite time.*

3 Heat kernel estimates

In order to apply the Feynman-Kac formula to the study of equations of the form (1.3) we need to study the transition function of the Bessel semigroup.

Let $\mathbb{P}_\mu$ denote the law of the Bessel process $(R_t)_{t\in\mathbb{R}_+}$ with parameter $\mu \in \mathbb{R}$, generator

$$L_\mu := \frac{1}{2}\frac{\partial^2}{\partial x^2} + \frac{2\mu+1}{2x}\frac{\partial}{\partial x}, \qquad x > 0,$$

and transition density $q_t^\mu(x,y)$, given when $\mu > -1$ by

$$q_t^\mu(x,y) = \frac{1}{t}\left(\frac{y}{x}\right)^\mu y e^{-\frac{x^2+y^2}{2t}} I_\mu\left(\frac{xy}{t}\right), \qquad x,y,t > 0,$$

where I_μ denotes the modified Bessel function of the first kind of order $\mu > -1$, cf. e.g. [3], Theorem 9.1.

The measure $\mathbb{P}_\mu$ is a probability measure on the filtered space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\in\mathbb{R}_+})$, where Ω is the space $C(\mathbb{R}_+, \mathbb{R}_+)$ of nonnegative continuous functions on $\mathbb{R}_+$, $\mathcal{F} = \sigma\{R_s,\, s \geq 0\}$ and $\mathcal{F}_t = \sigma\{R_s,\, 0 \leq s \leq t\}$, $t \geq 0$. Here $R_s(\omega) = \omega(s)$ for all $s \in \mathbb{R}_+$ and $\omega \in \Omega$. Recall that when

$$d := 2 + 2\mu$$

is a positive integer we have $R_t = \|W_t\|$ under $\mathbb{P}_\mu$, where $(W_t)_{t\in\mathbb{R}_+}$ is a standard Brownian motion in $\mathbb{R}^d$, and $(R_t)_{t\in\mathbb{R}_+}$ solves the stochastic differential equation

$$dR_t = \frac{2\mu+1}{2}\frac{dt}{R_t} + dB_t,$$

where $(B_t)_{t\in\mathbb{R}_+}$ is a one-dimensional standard Brownian motion. Moreover, for $\mu, \nu \in \mathbb{R}$ and F any $\mathcal{F}_t$-measurable positive random variable, we have

$$\begin{aligned} &\mathbb{E}_\mu\left[\left(\frac{x}{R_t}\right)^\mu F \exp\left(-\frac{\nu^2}{2}\int_0^t \frac{ds}{R_s^2}\right)\mathbf{1}_{\{t<\tau_0\}}\Big| R_0 = x\right] \\ &\quad = \mathbb{E}_\nu\left[\left(\frac{x}{R_t}\right)^\nu F \exp\left(-\frac{\mu^2}{2}\int_0^t \frac{ds}{R_s^2}\right)\mathbf{1}_{\{t<\tau_0\}}\Big| R_0 = x\right], \qquad x \geq 0, \end{aligned} \tag{3.1}$$

where $\mathbb{E}_\mu$ denotes the expectation under $\mathbb{P}_\mu$ and τ_0 denotes the first time $(R_t)_{t\in\mathbb{R}_+}$ reaches 0, cf. [12] Lemma (4.5), [10] Chapter XI, Exercise 1.22, and [13] Chapter 6, § 2.2.

For any $a \geq 0$ let $p_t^a(x,y)$, $t > 0$, denote the transition densities of the Markov process $(X_t)_{t\in\mathbb{R}_+}$ with generator $L_\mu - V$, where $V(x) = a/x^2$, $x > 0$. Recall that from the Feynman-Kac formula we have

$$p_t^a(x,y) = q_t^\mu(x,y)\,\mathbb{E}_\mu\left[\exp\left(-a\int_0^t \frac{1}{R_s^2}\,ds\right)\Big| R_t = y,\ R_0 = x\right], \qquad x,y \geq 0, \tag{3.2}$$

for all $\mu \in \mathbb{R}$ and $a \geq 0$. On the other hand, letting

$$\tau_0 = \inf\{t > 0 \ : \ R_t = 0\}$$

denote the first hitting time of 0 by $(R_t)_{t\in\mathbb{R}_+}$, it is known that $\tau_0 = +\infty$ when $\mu \geq 0$, and the integral $\displaystyle\int_0^t \frac{ds}{R_s^2}$ diverges a.s. when $\mu < 0$, cf. [12] and [13] Chapter 6. Hence we also have

$$p_t^a(x,y) = q_t^\mu(x,y)\,\mathbb{E}_\mu\left[\exp\left(-a\int_0^t \frac{1}{R_s^2}\,ds\right)\mathbf{1}_{\{t<\tau_0\}}\Big|R_t = y,\ R_0 = x\right], \qquad x,y \geq 0,$$

for all $\mu \in \mathbb{R}$ and $a > 0$. Note that in general, (3.2) yields

$$\lim_{a\searrow 0} p_t^a(x,y) = p_t^0(x,y)\mathbb{P}_\mu(t < \tau_0),$$

thus, when $\mu < 0$, $p_t^a(x,y)$ does not converge to $p_t^0(x,y)$ as a tends to 0.

For any $\mu \in \mathbb{R}$ and $a \geq 0$, let

$$\nu := \begin{cases} \sqrt{\mu^2 + 2a}, & \text{if } a > 0, \\ \\ \mu, & \text{if } a = 0, \end{cases}$$

and

$$n := \frac{\nu - \mu}{2} = \begin{cases} \dfrac{\sqrt{\mu^2 + 2a} - \mu}{2}, & \text{if } a > 0, \\ \\ 0, & \text{if } a = 0. \end{cases} \tag{3.3}$$

Moreover, when $a = 0$ we will assume that $\nu = \mu > -1$.

Lemma 3.1 *For all $\mu \in \mathbb{R}$ and $a \geq 0$ we have*

$$p_t^a(x,y) = x^{2n}y^{-2n}q_t^\nu(x,y), \qquad x,y,t \geq 0. \tag{3.4}$$

Proof. Clearly it suffices to consider the case $a > 0$. By an application of (3.1) to

$$F := \left(\frac{x}{R_t}\right)^{-\mu} U \exp\left(\frac{\mu^2}{2}\int_0^t \frac{ds}{R_s^2}\right),$$

where U is an $\mathcal{F}_t$-measurable non-negative random variable, we get

$$\mathbb{E}_\mu\left[U\exp\left(-\frac{\nu^2-\mu^2}{2}\int_0^t \frac{ds}{R_s^2}\right)\mathbf{1}_{\{t<\tau_0\}}\Big|R_0 = x\right] = \mathbb{E}_\nu\left[\left(\frac{x}{R_t}\right)^{\nu-\mu} U\mathbf{1}_{\{t<\tau_0\}}\Big|R_0 = x\right], \tag{3.5}$$

hence

$$\begin{aligned}\mathbb{E}_\mu\left[\exp\left(-\frac{\nu^2-\mu^2}{2}\int_0^t\frac{ds}{R_s^2}\right)\mathbf{1}_{\{R_t\in dy\}}\mathbf{1}_{\{t<\tau_0\}}\Big|R_0=x\right]\\ =\ \mathbb{E}_\nu\left[\left(\frac{x}{R_t}\right)^{\nu-\mu}\mathbf{1}_{\{R_t\in dy\}}\mathbf{1}_{\{t<\tau_0\}}\Big|R_0=x\right]\\ =\ \left(\frac{x}{y}\right)^{\nu-\mu}\mathbb{P}_\nu(R_t\in dy\mid R_0=x),\end{aligned}$$

and

$$\mathbb{E}_\mu\left[\exp\left(-\frac{\nu^2-\mu^2}{2}\int_0^t\frac{ds}{R_s^2}\right)\mathbf{1}_{\{t<\tau_0\}}\Big|R_t=y,\ R_0=x\right]=\left(\frac{x}{y}\right)^{\nu-\mu}\frac{q_t^\nu(x,y)}{q_t^\mu(x,y)},$$

which yields (3.4) due to (3.2). □

Note that in case $\mu<-1$, letting $\tilde\nu=-\sqrt{\mu^2+2a}<-1$ and $\tilde n=(\tilde\nu-\mu)/2$, $a\geq 0$, the above argument would yield the upper bound

$$p_t^a(x,y)\leq x^{2\tilde n}y^{-2\tilde n}q_t^{\tilde\nu}(x,y),\qquad x,y,t\geq 0,\tag{3.6}$$

which is weaker than (3.4) but remains valid in the limit as a tends to 0.

From Lemma 3.1 we deduce the following lower bounds. In the sequel, $c>0$ denotes a generic positive constant whose value depends on the context.

Lemma 3.2 *Let $\mu\in\mathbb{R}$ and $a>0$. For all sufficiently large $t>t_0$ we have*

$$p_t^a(x,y)\geq ct^{-\nu-1}x^{2n}y^{2n+d-1}\mathbf{1}_{[0,\sqrt{t}]}(x)\mathbf{1}_{[0,\sqrt{t}]}(y),\qquad x,y\geq 0.\tag{3.7}$$

Proof. From the equivalence

$$I_\nu(z)\simeq cz^\nu$$

as z tends to 0, which is valid for $\nu>-1$, we get

$$q_t^\nu(x,y)\simeq ct^{-\nu-1}y^{2\nu+1}e^{-\frac{x^2+y^2}{2t}},\qquad\text{as } t\to\infty,$$

which, due to Lemma 3.1, shows that for all $x,y\geq 0$ we have

$$p_t^a(x,y)\simeq ct^{-2n-d/2}x^{2n}y^{2n+d-1}e^{-\frac{x^2+y^2}{2t}},\qquad\text{as } t\to\infty,\tag{3.8}$$

hence (3.7) holds. □

When $\mu > -1$ and $a = 0$, an argument similar to that of Lemma 3.2 yields

$$p_t^0(x,y) \geq ct^{-\mu-1}y^{d-1}\mathbf{1}_{[0,\sqrt{t}]}(x)\mathbf{1}_{[0,\sqrt{t}]}(y).$$

As a consequence of Lemma 3.2, for all sufficiently large $t > 0$ we have for all $x, y \geq 0$,

$$p_t^a(x,y) \geq ct^{-\nu-1}y^{2n+d-1}\mathbf{1}_{[\alpha,\sqrt{t}]}(x)\mathbf{1}_{[0,\sqrt{t}]}(y), \tag{3.9}$$

and

$$p_t^a(x,y) \geq ct^{-\nu-1}x^{2n}\mathbf{1}_{[0,\sqrt{t}]}(x)\mathbf{1}_{[\alpha,\sqrt{t}]}(y). \tag{3.10}$$

The next lemma provides upper bounds for the transition densities $p_t^a(x,y)$.

Lemma 3.3 *Let $\mu \in \mathbb{R}$ and $a > 0$. There exists $t_0 > 0$ such that for all $t > t_0$ we have*

$$p_t^a(x,y) \leq ct^{-n-d/2}y^{2n+d-1}, \qquad x, y \geq 0. \tag{3.11}$$

Proof. Due to (3.8) and the fact that $x^{2n}e^{-\frac{x^2}{2t}} \leq Ct^n$ for all $x \geq 0$, where $C > 0$ is a constant, we have

$$\begin{aligned} p_t^a(x,y) &\leq ct^{-2n-d/2}x^{2n}y^{2n+d-1}e^{-\frac{x^2}{2t}} \\ &\leq ct^{-n-d/2}y^{2n+d-1}, \qquad x, y \geq 0. \end{aligned}$$

□

When $\mu > -1$ and $a = 0$, by the argument leading to (3.11) we get

$$p_t^0(x,y) = q_t^\mu(x,y) \leq ct^{-d/2}y^{d-1}, \qquad x, y \geq 0. \tag{3.12}$$

On the other hand, when $\mu < 0$, using (3.6) yields another upper bound

$$p_t^a(x,y) \leq ct^{-\tilde{n}-d/2}y^{2\tilde{n}+d-1}, \qquad x, y \geq 0,$$

with $\tilde{\nu} = -\sqrt{\mu^2 + 2a} > -1$, $a \geq 0$, and $\tilde{n} = (\tilde{\nu} - \mu)/2$, which is not directly comparable with (3.11) but, unlike (3.11), recovers (3.12) as a tends to 0, and conducts to (6.7) below.

4 Semigroup bounds

In this section, from the heat kernel bounds of the previous section, we derive the semigroup bounds that will be used to prove the results of Sections 5 and 6.

In the next lemma we give a lower bound for the semigroup $(T_t^a)_{t\geq 0}$ generated by $L_\mu - V$, i.e.

$$T_t^a \phi(x) = \int_0^\infty \phi(y) p_t^a(x,y)\, dy, \tag{4.1}$$

is the solution $f_t(x)$ of the linear equation

$$\begin{cases} \dfrac{\partial f_t}{\partial t}(x) = \dfrac{1}{2}\Delta f_t(x) + \dfrac{2\mu+1}{2x}\nabla f_t(x) - \dfrac{a}{x^2} f_t(x), \quad t > 0, \\ \\ f_0(x) = \phi(x), \qquad x > 0. \end{cases}$$

Without loss of generality we assume that $\phi > 0$ a.e. on a bounded interval $(s,t) \subset \mathbb{R}_+$, and that $\int_0^\infty \phi(y)\, dy < \infty$. The next lemma uses the number n defined in (3.3).

Lemma 4.1 *Assume that $\mu \in \mathbb{R}$ and $a > 0$. Then for all sufficiently large $t > 1$ there holds*

$$T_t^a \phi(x) \geq c t^{-2n-d/2} x^{2n} \mathbf{1}_{[0,\sqrt{t}]}(x), \qquad x \geq 0. \tag{4.2}$$

Proof. We use the bound (3.10), which yields

$$\begin{aligned} f_t(x) &= \int_0^\infty \phi(y) p_t^a(x,y)\, dy \\ &\geq c t^{-2n-d/2} x^{2n} \mathbf{1}_{[0,\sqrt{t}]}(x) \int_\alpha^{\sqrt{t}} \phi(y)\, dy \\ &\geq c' t^{-2n-d/2} x^{2n} \mathbf{1}_{[0,\sqrt{t}]}(x) \int_0^\infty \phi(y)\, dy, \qquad x \geq 0, \end{aligned}$$

for all sufficiently large t, due to our assumptions on $\phi(y)$. □

Note that the bound (4.2) is also valid if $\mu > -1$ and $a = 0$, in which case it reads

$$T_t^0 \phi(x) \geq c t^{-d/2} \mathbf{1}_{[0,\sqrt{t}]}(x), \qquad x \geq 0. \tag{4.3}$$

Let $(X_t)_{t\in\mathbb{R}_+}$ be the Markov process with generator $L_\mu - V$.

Lemma 4.2 *Assume that $\mu \in \mathbb{R}$ and $a > 0$. Let $\alpha \in (0,1)$ and $x \in [\alpha, 1]$. For all t large enough, all $s \in [1, t/2]$ and all $y \in [0, \sqrt{t-s}]$, there holds*

$$\mathbb{E}\left[f_{t-s}^\beta(X_s) \mathbf{1}_{[0,\sqrt{t-s}]}(X_s) \middle| X_t = y, X_0 = x \right] \geq c t^{-(n+2\beta n+\beta d/2)} s^{\beta n}. \tag{4.4}$$

Proof. From Lemma 4.1 above, for all t large enough we have

$$f_{t-s}(X_s) \geq c(t-s)^{-2n-d/2} X_s^{2n} \mathbf{1}_{[0,\sqrt{t-s}]}(X_s), \qquad 0 \leq s \leq t, \tag{4.5}$$

and therefore,

$$\begin{aligned}\mathbb{E}\left[f_{t-s}^{\beta}(X_s)\mathbf{1}_{[0,\sqrt{t-s}]}(X_s)\Big| X_t = y, X_0 = x\right] \\ \geq \ c(t-s)^{-\beta(2n+d/2)}\,\mathbb{E}\left[X_s^{2\beta n}\mathbf{1}_{[0,\sqrt{t-s}]}(X_s)\Big| X_t = y, X_0 = x\right].\end{aligned} \tag{4.6}$$

We now proceed to obtain a lower bound for the moments of the process $(X_t)_{t\in\mathbb{R}_+}$. Let $\alpha \in (0,1)$ be given, and let $x \in [\alpha, 1]$. Due to (3.9) we have, for all sufficiently large $t > 2$ and all $1 \leq s \leq t/2$,

$$p_s^a(x,z) \geq cs^{-2n-d/2}z^{2n+d-1}\mathbf{1}_{[0,\sqrt{s}]}(z), \qquad x, y \geq 0,$$

and

$$p_{t-s}^a(z,y) \geq c(t-s)^{-2n-d/2}z^{2n}y^{2n+d-1}\mathbf{1}_{[0,\sqrt{t-s}]}(z)\mathbf{1}_{[0,\sqrt{t-s}]}(y), \qquad x, y \geq 0.$$

Together with Lemma 3.3, the above inequalities render

$$\frac{p_s^a(x,z)p_{t-s}^a(z,y)}{p_t^a(x,y)} \geq cz^{4n+d-1}\frac{s^{-2n-d/2}(t-s)^{-2n-d/2}}{t^{-n-d/2}}\mathbf{1}_{[0,\sqrt{s}]}(z), \qquad x, y, z \geq 0.$$

It follows that

$$\begin{aligned}&\mathbb{E}\left[X_s^{2\beta n}\mathbf{1}_{[0,\sqrt{t-s}]}(X_s)\Big| X_t = y,\ X_0 = x\right] \\ &= \int_0^{\infty} z^{2\beta n}\mathbf{1}_{[0,\sqrt{t-s}]}(z)\mathbb{P}(X_s \in dz \mid X_t = y, X_0 = x) \\ &\geq \ ct^{n+d/2}s^{-(2n+d/2)}(t-s)^{-(2n+d/2)}\int_0^{\sqrt{s}} z^{4n+d-1+2\beta n}\,dz \\ &= \ ct^{n+d/2}(t-s)^{-(2n+d/2)}s^{\beta n}.\end{aligned}$$

□

If $\mu > -1$ and $a = 0$, by (3.12) and the argument of Lemma 4.2 we have

$$\mathbb{E}\left[\mathbf{1}_{[0,\sqrt{t-s}]}(X_s)f_{t-s}^{\beta}(X_s)\Big| X_t = y, X_0 = x\right] \ \geq \ ct^{-\beta d/2} \tag{4.7}$$

under the same conditions as in (4.4). Next we state an estimate which will be useful in Section 6, and is valid as above under the condition

$$(\mu, a) \in (\mathbb{R}\times(0,\infty)) \cup ((-1,\infty)\times\{0\}).$$

Proposition 4.3 *Assume that* $\phi \geq 0$ *and*

$$\int_0^\infty y^{2n+d-1}\phi(y)\,dy < \infty.$$

Then there exists $t_0 > 0$ *such that for all* $t > t_0$ *we have*

$$\|T_t^a\phi\|_{L^\infty(\mathbb{R})} \leq ct^{-n-d/2}.$$

Proof. From Lemma 3.3 we get

$$\begin{aligned} T_t^a\phi(x) &= \int_0^\infty \phi(y)p_t^a(x,y)\,dy \\ &\leq ct^{-n-d/2}\int_0^\infty y^{2n+d-1}\phi(y)\,dy, \qquad x \in \mathbb{R}. \end{aligned}$$

□

5 Explosion via the Feynman-Kac formula

Let now $g_t(x)$ denote the mild solution to the semilinear equation

$$\begin{cases} \dfrac{\partial g_t}{\partial t}(x) = \dfrac{1}{2}\dfrac{\partial^2 g_t}{\partial x^2}(x) + \dfrac{2\mu+1}{2x}\dfrac{\partial g_t}{\partial x}(x) - \dfrac{a}{x^2}g_t(x) + g_t(x)f_t^\beta(x), \quad t > 0, \\ \\ g_0(x) = \phi(x), \qquad x > 0, \end{cases} \tag{5.1}$$

where $\mu \in \mathbb{R}$, $a \geq 0$ and $f_t(x)$ is defined in (4.1).

Proposition 5.1 *Let* $\mu \in \mathbb{R}$, $a > 0$ *and* $\alpha \in (0,1]$. *There exists* $c_3, c_4 > 0$ *such that for all* $x > 0$ *and all sufficiently large* $t \geq 1$,

$$g_t(x) \geq c_4 \exp\left(c_3 t^{1-\beta(n+d/2)-n}\right)\mathbf{1}_{[\alpha,1]}(x).$$

Proof. The Feynman-Kac representation of (5.1) yields

$$g_t(x) = \int_0^\infty \phi(y)p_t^a(x,y)\,\mathbb{E}\left[\exp\int_0^t f_{t-s}^\beta(X_s)\,ds \,\middle|\, X_t = y,\ X_0 = x\right]dy.$$

Let $x \in [\alpha, 1]$. Using Jensen's inequality we obtain, for all t large enough,

$$\begin{aligned} g_t(x) &\geq \int_0^\infty \phi(y)p_t^a(x,y)\exp\left(c_1\int_1^{t/2}\mathbb{E}\left[f_{t-s}^\beta(X_s) \mid X_t = y, X_0 = x\right]ds\right)dy \\ &\geq \int_\alpha^1 \phi(y)p_t^a(x,y)\exp\left(c_2 t^{-(n+2\beta n+\beta d/2)}\int_1^{t/2} s^{\beta n}\,ds\right)dy \\ &\geq c_4 t^{-(1+\nu)}\exp\left(c_3 t^{1-\beta(n+d/2)-n}\right), \qquad x \geq 0, \end{aligned}$$

for some positive constants $c_1, c_2, c_3 > 0$, where we used Lemma 4.2 to obtain the second inequality. □

Notice that, when $\mu > -1$ and $a = 0$, the above argument together with (4.7) gives

$$g_t(x) \geq c_4 \exp\left(c_3 t^{1-\beta d/2}\right) \mathbf{1}_{[\alpha,1]}(x), \qquad x \geq 0.$$

As a consequence of the above proposition, g grows to $+\infty$ uniformly on $[\alpha, 1]$ provided that $1-\beta(n+d/2)-n > 0$, and this implies the following result, in which n is defined by (3.3).

Theorem 5.2 *Let $\mu \in \mathbb{R}$, $a > 0$, and assume that*

$$\beta < \frac{1-n}{n+d/2}.$$

Then the mild solution $u_t(x)$ of (1.3) blows up in finite time.

Proof. Let $u_t(x)$ denote the solution of (1.3). Since $g_t(x) \leq u_t(x)$, Proposition 5.1 implies that $u_t(x)$ grows to $+\infty$ uniformly on $[\alpha, 1]$ as $t \to \infty$. According to a well-known argument [4], this is sufficient to prove explosion in finite time of $u_t(x)$. Indeed, let $t_0 \geq 1$, $\tilde{u}_t = u_{t+t_0}$ and $K(t_0) = \min_{x\in[\alpha,1]} u_{t_0}(x)$. Then $\tilde{u}_t$ solves

$$\tilde{u}_t(x) = \int p_t^a(x,y) u_{t_0}(y)\, dy + \int_0^t \int p_{t-s}^a(x,y) \left(\tilde{u}_s(y)\right)^{1+\beta} dy\, ds,$$

hence

$$\min_{x\in[\alpha,1]} \tilde{u}_t(x) \geq \xi K(t_0) + \xi \int_0^t \left(\min_{x\in[\alpha,1]} \tilde{u}_s(x)\right)^{1+\beta} ds, \quad t \in [0,1],$$

where

$$\xi := \min_{r\in[0,1]} \min_{x\in[\alpha,1]} \mathbb{P}_x(X_r \in [\alpha,1]). \tag{5.2}$$

From Lemma 3.1 it follows that the function

$$(r,x) \mapsto P(X_r \in [\alpha,1] \mid X_0 = x) = \int_\alpha^1 p_r^a(x,y)\, dy = x^{2n} \int_\alpha^1 y^{-2n} q_r^\nu(x,y)\, dy$$

is continuous and strictly positive on $[0,1] \times [\alpha,1]$. Therefore $\xi > 0$, and it suffices to choose $t_0 > 0$ sufficiently large so that the blow-up time ρ_0 of the equation

$$y(t) = \xi K(t_0) + \xi \int_0^t y^{1+\beta}(s)\, ds$$

is smaller than 1 to conclude that $u_t(x)$ blows up at time $t_0 + \rho_0$. □

We remark that in case $\mu > -1$ and $a = 0$, the conclusion of Theorem 5.2 holds for

$$\beta < 2/d.$$

The next result holds for

$$(\mu, a) \in (\mathbb{R} \times (0,\infty)) \cup ((-1,\infty) \times \{0\}).$$

Corollary 5.3 *Let $G : \mathbb{R}_+ \to \mathbb{R}_+$ be increasing and convex, such that*

$$\frac{G(z)}{z} \sim \kappa_1 z^\beta \text{ as } z \to 0, \tag{5.3}$$

for some $\kappa_1 > 0$, and let $w : \mathbb{R}_+ \times \mathbb{R}^d \to \mathbb{R}_+$ be a measurable function satisfying

$$w_t(x) \geq \kappa_2 t^\zeta \mathbf{1}_{(0,1)}(t^{-1/2}x) \tag{5.4}$$

for all $t \geq 1$ and some $\kappa_2 > 0$. Then any nontrivial positive solution of the semilinear equation

$$\begin{cases} \dfrac{\partial u_t}{\partial t}(x) = \dfrac{1}{2}\dfrac{\partial^2 u_t}{\partial x^2}(x) + \dfrac{2\mu+1}{2x}\dfrac{\partial u_t}{\partial x}(x) - \dfrac{a}{x^2}u_t(x) + w_t(x)G(u_t(x)), \\ u_0(x) = \phi(x) \geq 0, \qquad x > 0, \quad t > 0, \end{cases} \tag{5.5}$$

blows up in finite time provided

$$\beta < \frac{1+\zeta-n}{n+d/2}.$$

Proof. The Feynman-Kac representation of (5.5) yields

$$u_t(x) = \int_0^\infty \phi(y) p_t^a(x,y)\, \mathbb{E}\left[\exp\left(\int_0^t w_{t-s}(X_s)\frac{G(u_{t-s}(X_s))}{u_{t-s}(X_s)}ds\right)\Big|\, X_t = y,\ X_0 = x\right] dy.$$

Since by assumption $w_{t-s}(X_s)G(u_{t-s}(X_s))/u_{t-s}(X_s)$ can be bounded from below by

$$\kappa_1\kappa_2(t-s)^\zeta \mathbf{1}_{[0,\sqrt{t-s}]}(X_s)u_{t-s}^\beta(X_s) \geq \kappa_1\kappa_2(t-s)^\zeta \mathbf{1}_{[0,\sqrt{t-s}]}(X_s)f_{t-s}^\beta(X_s),$$

we get, using Lemma 4.2 as in the proof of Proposition 5.1, that

$$\begin{aligned} u_t(x) \geq\ & \int_0^\infty \phi(y)p_t^a(x,y)\,\mathbb{E}\left[\exp\left(\kappa_1\kappa_2\int_0^t (t-s)^\zeta \mathbf{1}_{[0,\sqrt{t-s}]}(X_s)f_{t-s}^\beta(X_s)ds\right)\Big|\, X_t = y,\ X_0 = x\right] dy \\ \geq\ & \int_0^\infty \phi(y)p_t^a(x,y)\exp\left(\kappa_1\kappa_2\int_0^t (t-s)^\zeta\,\mathbb{E}\left[\mathbf{1}_{[0,\sqrt{t-s}]}(X_s)f_{t-s}^\beta(X_s)\Big|\, X_t = y,\ X_0 = x\right] ds\right) dy \\ \geq\ & \int_\alpha^1 \phi(y)p_t^a(x,y)\exp\left(c_2 t^{-(n+2\beta n+\beta d/2)}\int_1^{t/2} s^{\zeta+\beta n}\,ds\right) dy \\ \geq\ & c_5 t^{-1-\nu}\exp\left(c_6 t^{1+\zeta-\beta(n+d/2)-n}\right) \end{aligned}$$

for all $x \in [\alpha, 1]$, all t large enough, and some constants $c_5, c_6 > 0$. Therefore,

$$\lim_{t\to\infty}\inf_{x\in[\alpha,1]} u_t(x) = \infty$$

due to the condition $1 + \zeta > \beta(n + d/2) + n$. The assertions follow in the same way as in the proof of Theorem 5.2. □

Again when $\mu > -1$ and $a = 0$, the conclusion of Corollary 5.3 also holds when

$$\beta < \frac{2(1+\zeta)}{d}.$$

6 Existence of global solutions

The following result gives conditions for existence of a nontrivial positive global solution. Its proof is very similar to that of Theorem 4.1 in [6], and is therefore omitted.

Theorem 6.1 *Let $\mu \in \mathbb{R}$ and $a \geq 0$, and consider the semilinear equation*

$$\begin{cases} \dfrac{\partial u_t}{\partial t}(x) = \dfrac{1}{2}\dfrac{\partial^2 u_t}{\partial x^2}(x) + \dfrac{2\mu+1}{2x}\dfrac{\partial u_t}{\partial x}(x) - \dfrac{a}{x^2}u_t(x) + t^\zeta G(u_t(x)), \quad t > 0, \\ \\ u_0(x) = \phi(x), \quad x > 0, \end{cases} \tag{6.1}$$

where $\zeta \in \mathbb{R}$, ϕ is bounded and measurable, and $G : \mathbb{R}_+ \to \mathbb{R}_+$ is a measurable function satisfying

$$0 \leq \frac{G(z)}{z} \leq \lambda z^\beta, \qquad z > 0, \tag{6.2}$$

for some $\lambda, \beta > 0$. If

$$\lambda\beta \int_0^\infty r^\zeta \|T_r^a \phi\|_{L^\infty(\mathbb{R})}^\beta \, dr < 1, \tag{6.3}$$

then (6.1) *admits a global solution.*

Notice that by choosing $\|\phi\|_{L^\infty(\mathbb{R})} > 0$ sufficiently small, it is possible to prove existence of a positive global solution of (6.1) under (6.3) and the less restrictive condition

$$0 \leq \frac{G(z)}{z} \leq \lambda z^\beta, \qquad z \in (0, c),$$

for some $\lambda, \beta, c > 0$, see [7], Theorem 4.1.

As a consequence of Theorem 6.1, an existence result can be obtained under an integrability condition on ϕ.

Theorem 6.2 *Given $(\mu, a) \in \mathbb{R} \times (0, \infty) \cup (-1, \infty) \times \{0\}$, let $G : \mathbb{R}_+ \to \mathbb{R}_+$ and $w : \mathbb{R}_+ \times \mathbb{R} \to \mathbb{R}_+$ be measurable functions such that*

$$G(z) \leq \kappa_1 z^{1+\beta}, \qquad z > 0, \quad \textit{and} \quad w_t(x) \leq \kappa_2 t^\zeta, \qquad (t, x) \in \mathbb{R}_+ \times \mathbb{R},$$

where $\beta, \kappa_1, \kappa_2 > 0$ *and* $\zeta \in \mathbb{R}$. *The equation*

$$\begin{cases} \dfrac{\partial u_t}{\partial t}(x) = \dfrac{1}{2}\dfrac{\partial^2 u_t}{\partial x^2}(x) + \dfrac{2\mu+1}{2x}\dfrac{\partial u_t}{\partial x}(x) - \dfrac{a}{x^2}u_t(x) + w_t(x)G(u_t(x)), \quad t > 0, \\ \\ u_0(x) = \phi(x), \quad x > 0, \end{cases} \tag{6.4}$$

admits a global solution on $(0, \infty)$ *provided*

$$\beta > \frac{1+\zeta}{n + d/2}. \tag{6.5}$$

Proof. Clearly, it suffices to consider the semilinear equation

$$\begin{cases} \dfrac{\partial u_t}{\partial t}(x) = \dfrac{1}{2}\dfrac{\partial^2 u_t}{\partial x^2}(x) + \dfrac{2\mu+1}{2x}\dfrac{\partial u_t}{\partial x}(x) - \dfrac{a}{x^2}u_t(x) + \kappa t^{\zeta} u_t^{1+\beta}(x), \quad t > 0, \\ \\ u_0(x) = \phi(x), \quad x > 0, \end{cases} \tag{6.6}$$

for a suitable constant $\kappa > 0$, and to apply Proposition 4.3 and Theorem 6.1. □

Let us remark that from (3.6) one can also show that when $\mu < -1$, $a \geq 0$, and $\mu^2 + 2a < 1$, Equation (6.4) admits a global solution on $(0, \infty)$ provided

$$\beta > \frac{2 + 2\zeta}{2 + \mu - \sqrt{\mu^2 + 2a}}. \tag{6.7}$$

The above bound (6.7) recovers the critical exponent $(1+\zeta)/(1+\mu)$ when a tends to 0, however it is weaker than (6.5).

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